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Sensitivity analysis of leaky-mode dispersion curves for horizontal stratified models

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SUMMARY

The leaky-mode dispersion extracted from seismograms and noise cross-correlation functions has gained lots of attention in recent years. It has been reported that leaky modes can provide constraints for subsurface structures, especially for *P*-wave velocities, which may compensate for the limits of traditional surface wave methods. For stable and reliable dispersion-curve inversion, the quantitative analysis of leaky-mode sensitivity is of great importance, which, however, has rarely been studied systematically. Limited by the forward modelling methods, the previous methods for calculating leaky-mode sensitivity are usually hindered by issues of mode skipping, low efficiency, etc. To this end, we propose an effective method that can calculate the leaky-mode sensitivity for various types of models based on the previously proposed forward modelling method named the semi-analytical spectral element method (SASEM). Using the intermediate results of SASEM, we derive analytical expressions for the sensitivity kernels with only matrix operations, which endows the sensitivity calculation procedure with high accuracy and reliability identical to the SASEM. In addition, we suggest a novel modal classification scheme to distinguish different kinds of leaky modes based on the sensitivity features. This scheme facilitates the stable identification of the most attractive guided-*P* modes from numerous normal and leaky modes, which removes obstacles in the dispersion-curve inversion using guided-*P* modes to constrain *P*-wave velocities. Several numerical tests are performed to demonstrate the high accuracy of the sensitivity calculation method and the effectiveness of the modal classification method. To assess the roles of leaky modes in the retrieval of underground structures, we perform comprehensive sensitivity analyses of leaky modes using both crust-scale and near-surface models. Besides the general conclusion that the joint inversion using normal and leaky modes can effectively retrieve *P*- and *S*-wave velocities, the feasibility of constraining models with the apparent Σ modes and the split guided-*P* mode dispersion curves has been demonstrated.

Key words: Guided waves; Surface waves and free oscillations; Theoretical seismology.

1 INTRODUCTION

Dispersion curves play an important role in characterizing seismic waves propagating along a stratified model and retrieving subsurface structures. The basis of dispersion curves is the mode theory of the horizontal stratified models (Aki & Richards 2002; Jensen *et al.* 2011). The surface-wave dispersion is typically described by normal modes that have been studied extensively in past years (e.g. Oliver & Ewing 1958). In addition to surface waves, leaky waves that propagate faster and transfer energy into the underlying half-space contribute significantly to the observed seismograms as well (Oliver & Major 1960). To analyse leaky waves, the theories of leaky modes are developed (e.g. Rosenbaum 1960). For elastic models, the normal modes correspond to the real roots of the dispersion functions while the leaky modes correspond to the complex roots. Seismograms were proven to be the comprehensive contributions of normal and leaky modes (Haddon 1984). In recent years, with the development of array-based observation technologies and modern dispersion analysis tools such as the Frequency-Bessel transform (Forbriger 2003; Wang *et al.* 2019), the leaky-mode dispersion extracted from seismograms and ambient noises were frequently reported and attracted a lot of attention because leaky modes provide model constraints that cannot be accessed easily from surface waves (Li *et al.* 2021, 2022; Sun *et al.* 2021). For example, a subset of

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leaky modes is primarily related to *P* waves (Su & Dorman 1965; Cochran *et al.* 1970) and has been proven effective for the retrieval of *P*-wave velocities (Boiero *et al.* 2013; Li *et al.* 2022; Wang *et al.* 2024).

With the widespread study and application of surface wave dispersion (e.g. Dorman & Ewing 1962; Xia *et al.* 1999; Wu *et al.* 2020), geoscientists are now very familiar with normal modes. Based on thorough analyses of the normal-mode sensitivity (partial derivatives of dispersion curves to model parameters), stable and reliable inversion schemes have been proposed to constrain the subsurface *S*-wave velocity structures (Xia *et al.* 2003; Pan *et al.* 2019). The sensitivity analysis serves as the foundation for the subsequent inversion mainly for two reasons. First, the sensitivity analysis assesses the potential of different modes to constrain the subsurface structures and can provide significant guides for successful inversion, including determining the model parameters to be retrieved and evaluating the reliability of inversion results (Pan *et al.* 2019). Secondly, sensitivity calculation is the core of the popular local linearization inversion that matches the theoretical dispersion curves with the observational ones. Compared with normal modes, although several attempts that estimate velocity structures using leaky modes have been reported in past years (e.g. Li *et al.* 2022; Shi *et al.* 2023; Wang *et al.* 2024; Zhang *et al.* 2024), quantitative studies on sensitivity calculation and analysis for different models remain scarce.

The calculation of normal-mode sensitivity is a classical problem in seismology and can be solved via different methods. A simple and well-known method to calculate the accurate sensitivity for normal modes was derived based on the variational method (Jeffreys 1961; Aki & Richards 2002). A fundamental assumption of the variational method is that the eigen wavefields should be integrable, which is valid for normal modes whose eigen wavefields decay exponentially with depth in the half-space. Unfortunately, the eigen wavefields for leaky modes have been reported to grow exponentially in the half-space (e.g. Gilbert 1964; Alsop *et al.* 1974; Haddon 1984; Jensen *et al.* 2011), which leads to the failure of the variational method. As an alternative, the finite-difference method (Dorman & Ewing 1962; Feng *et al.* 2005) appears to be a straightforward approach for calculating the leaky-mode sensitivities. The main drawback of the finite-difference method is the huge time-cost since calculating the sensitivity of each mode requires twice forward-modelling processes. With the forward modelling method based on searching roots of the dispersion function, the sensitivity can be calculated with the differentiation of the dispersion function (Cochran *et al.* 1970; Wang *et al.* 2024). However, due to the dependence on initial guesses, the forward modelling based on root searching may result in mode skipping near the osculation points (or mode-kissing points). To circumvent the challenge of determining the complex roots of dispersion functions, Li *et al.* (2021) determined dispersion curves by tracing peaks of the dispersion spectra, which, however, costs more computation and requires careful numerical processes for modal tracing. Lin *et al.* (2023) suggested using the global stiffness matrix, which is directly related to Green's functions. Then the leaky modes can be captured by searching the local minima of the determinant of the global stiffness matrix. With a spectral-element method based on the perfectly matched layer (PML), Shi *et al.* (2023) analysed the sensitivity of leaky modes propagating in shallow marine sediments that exhibit simple leaky-mode dispersion. The success of the PML method hinges on designing appropriate PML parameters and spurious mode filters to eliminate Berenger modes originating from the artificial layer, which is sometimes difficult for models with complicated dispersion features. For practical applications in complex field data, the development of a stable and reliable sensitivity calculation method is urgent.

Unlike normal modes, another challenge in sensitivity analysis and inversion of leaky modes is the correct modal classification because of the complexity of leaky-mode dispersion. With quantitative analysis of the leaky modes on the (+, −) Riemann sheet, Cochran *et al.* (1970) divided these leaky modes into two groups: Π modes (dependent on *P*-wave velocities) and Σ modes (dependent on *S*-wave velocities). Similarly, Watson (1972) called the two groups of leaky modes on the (+, −) sheet the PL and organ-pipe modes. Because of the strong leakage (or attenuation), the vast majority of the Σ modes are invisible in field data, which makes it hard to link and compare the observed dispersion curves with the theoretical solutions unless the theoretical Σ modes are removed. However, the automatic classification of Π and Σ modes is challenging because they are always hybrid in the theoretical leaky-mode solutions. To our knowledge, it is impossible to calculate a specific type of mode separately. As a result, performing inversion to estimate *P*-wave velocities using leaky modes requires an indispensable classification procedure to identify the *P*-wave-dominated modes from the complete mode solutions. Because simply classifying leaky modes according to their attenuation coefficient may easily fail for complicated models with low Poisson's ratios or high (low) velocity layers (Wang *et al.* 2024), an effective, convenient, and stable modal classification technique is also required for further application in practical models.

To this end, this study develops a sensitivity analysis framework based on the previously proposed semi-analytical spectral element method (SASEM; Shi *et al.* 2022), which can calculate the sensitivity and perform modal classification for leaky modes. The main contents of this study are as follows. First, we briefly introduce the forward modelling method based on the improved SASEM and the algorithm for sensitivity calculation. Secondly, a strategy for modal classification is proposed, which allows us to analyse and use particular types of modes among the complicated leaky-mode solutions in inversion. The proposed method is verified by synthetic models with and without a low-velocity layer. Finally, we perform sensitivity analyses for the leaky modes of two more practical models to guide the dispersion curve inversion in crust-scale and near-surface investigations.

2 FORWARD MODELLING AND SENSITIVITY CALCULATION

2.1 SASEM for leaky-mode calculation

Forward modelling is essential for the sensitivity analysis and inversion of dispersion curves. For normal modes, a series of forward modelling methods have been developed (e.g. Haskell 1953; Knopoff 1964; Chen 1993). Based on these numerical methods, sensitivity can be measured efficiently and accurately. One of the main difficulties in leaky-mode sensitivity analysis is the calculation of dispersion curves. Calculating leaky-mode dispersion curves based on searching roots of the dispersion equations in the complex domain is usually not an easy task because of the probable occurrence of root skipping. In recent years, different kinds of discretization methods (Denolle *et al.* 2012; Haney & Tsai 2017; Hawkins 2018) become popular for calculating theoretical dispersion curves. These methods discretize the model

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using numerous nodes and determine different modes with eigenvalue decomposition, which can avoid the root-skipping problem. For leaky-mode problems, traditional discretization methods usually require well-designed artificial boundary conditions such as the PML (Treyssede *et al.* 2014; Shi *et al.* 2023). Otherwise, we must deal with the complicated nonlinear eigenvalue problem (Hayashi & Inoue 2014). Shi *et al.* (2022) proposed a novel and effective SASEM to calculate leaky modes located on the (+, −) Riemann sheet. The proposed SASEM can obtain the modal solutions stably and efficiently without any prior estimations or artificial boundary conditions, which is thus a good choice for leaky-mode inversion.

In this study, we suggest a modified forward modelling method based on the SASEM. Following the derivations in Shi *et al.* (2022), the wavefields in the finite-thickness layers of the stratified model (Fig. 1) are determined by the following frequency–wavenumber domain wave equations for wavefield displacements (e.g. Chen 1999):

$$\rho\omega^2 U_r(z) - k^2(\lambda+2\mu)U_r(z) - k\lambda\frac{\partial U_z(z)}{\partial z} - k\frac{\partial}{\partial z}[\mu U_z(z)] + \frac{\partial}{\partial z}\left[\mu\frac{\partial U_r(z)}{\partial z}\right] = 0, \tag{1}$$

$$\rho\omega^2 U_z(z) - k^2\mu U_z(z) + \mu k\frac{\partial U_r(z)}{\partial z} + k\frac{\partial}{\partial z}[\lambda U_r(z)] + \frac{\partial}{\partial z}\left[(\lambda+2\mu)\frac{\partial U_z(z)}{\partial z}\right] = 0, \tag{2}$$

where ω is the circular frequency, k is the horizontal wavenumber, λ and μ represent the Lamé parameters, ρ represents the density, $U_r(z)$ and $U_z(z)$ denote the horizontal and vertical displacements. When using α and β to represent the *P*- and *S*-wave velocities, the Lamé parameters are determined with $\lambda = \rho\alpha^2 - 2\rho\beta^2$ and $\mu = \rho\beta^2$.

The half-space wavefields are determined by wave potential equations. For both *P*- and *S*-wave potentials, their governing equations have similar forms to the acoustic wave equations and can be written as:

$$\rho\omega^2 p(z) - k^2(\lambda+2\mu)p(z) + (\lambda+2\mu)\frac{\partial^2 p(z)}{\partial z^2} = 0, \tag{3}$$

$$\rho\omega^2 q(z) - k^2\mu q(z) + \mu\frac{\partial^2 q(z)}{\partial z^2} = 0, \tag{4}$$

where $p(z)$ and $q(z)$ denote *P*- and *S*-wave potential functions respectively. In isotropic media, the relations between elastic displacements and potential functions are expressed using:

$$p(z) = \frac{\alpha^2}{\omega^2}\left(kU_r(z) + \frac{\partial U_z(z)}{\partial z}\right), \tag{5}$$

$$q(z) = -\frac{\beta^2}{\omega^2}\left(kU_z(z) + \frac{\partial U_r(z)}{\partial z}\right). \tag{6}$$

Besides the governing equations, the boundary conditions are essential to determine wavefields. For seismic problems, a free-surface boundary condition is usually adopted, which means the traction is zero at the surface of the model. In half-space, we impose a boundary condition with no reflected waves. For internal interfaces, all the elastic displacements and tractions are continuous while the potential functions are not.

With the upper wave equations and boundary conditions, SASEM converts the modal problem into matrix eigenvalue problems. Shi *et al.* (2022) determined leaky modes by solving a cubic eigenvalue problem. With the improved derivations shown in Appendix A, we have a simpler quadratic eigenvalue problem:

$$\omega^2\mathbf{M}\mathbf{U} - \mathbf{E}\mathbf{U} = \gamma_{N+1}^2\mathbf{K}_2\mathbf{U} + \gamma_{N+1}\mathbf{K}_1\mathbf{U}, \tag{7}$$

where $\mathbf{M}$, $\mathbf{E}$, $\mathbf{K}_1$ and $\mathbf{K}_2$ are the coefficient matrices determined by the model parameters, $\mathbf{U}$ is a column vector composed of the displacement and potential wavefields defined at different nodes, γ_{N+1} denotes the vertical wavenumber in the half-space. In the view of an eigenvalue problem, γ_{N+1} denotes the eigenvalue and $\mathbf{U}$ denotes the right eigenvectors. Compared with the cubic eigenvalue problem, the improved method results in easier mathematical processing and less computation amount. For eq. (7), applying a reduction approach gives a linear eigenvalue problem that can be directly solved by eigenvalue decomposition (see Appendix A).

When the half-space vertical wavenumber γ_{N+1} of a specific mode is obtained, the corresponding horizontal wavenumber k is calculated using $k = \sqrt{\omega^2/\beta_{N+1}^2 - \gamma_{N+1}^2}$, where β_{N+1} represents the *S*-wave velocity of the half-space. The real part of k affects the propagating velocity and the imaginary part of k represents the attenuation of waves in horizontal direction. The associated phase velocity v_c and attenuation coefficient σ of a given mode can be obtained using

$$v_c = \omega/\mathrm{Re}(k), \tag{8}$$

$$\sigma = -\mathrm{Im}(k). \tag{9}$$

2.2 Sensitivity calculation

2.2.1 *Partial derivatives*

The classical surface-wave dispersion inversion is a process of matching observed dispersion curves with the theoretical ones. The most widely used inversion methods are based on local linearization (Xia *et al.* 1999; Pan *et al.* 2019). The sensitivity of phase-velocity dispersion curves, which is usually measured using $\partial v_c/\partial m$ (m represents any parameter including the *P*/*S*-wave velocities and density), indicates how the model parameters affect phase velocities and is essential for local linearization inversion. This study proposes an effective method to calculate the partial derivatives with respect to the phase velocity under the framework of SASEM. The first step is to calculate the derivatives of eq. (7) to the model parameter m. Since both the coefficient matrices and eigenvector are functions with respect to m, we obtain the following differential equation:

$$\omega^2\frac{\partial\mathbf{M}}{\partial m}\mathbf{U} + \omega^2\mathbf{M}\frac{\partial\mathbf{U}}{\partial m} - \frac{\partial\mathbf{E}}{\partial m}\mathbf{U} - \mathbf{E}\frac{\partial\mathbf{U}}{\partial m} = 2\gamma_{N+1}\frac{\partial\gamma_{N+1}}{\partial m}\mathbf{K}_2\mathbf{U} + \gamma_{N+1}^2\frac{\partial\mathbf{K}_2}{\partial m}\mathbf{U} + \gamma_{N+1}^2\mathbf{K}_2\frac{\partial\mathbf{U}}{\partial m} + \frac{\partial\gamma_{N+1}}{\partial m}\mathbf{K}_1\mathbf{U} + \gamma_{N+1}\frac{\partial\mathbf{K}_1}{\partial m}\mathbf{U} + \gamma_{N+1}\mathbf{K}_1\frac{\partial\mathbf{U}}{\partial m}. \tag{10}$$

Before handling eq. (10), we introduce a new vector $\mathbf{W}$ that satisfies

$$\omega^2\mathbf{W}\mathbf{M} - \mathbf{W}\mathbf{E} = \gamma_{N+1}^2\mathbf{W}\mathbf{K}_2 + \gamma_{N+1}\mathbf{W}\mathbf{K}_1. \tag{11}$$

Because of the position of $\mathbf{W}$ in eq. (11), $\mathbf{W}$ is called the left eigenvector. The determination of $\mathbf{W}$ can be found in Appendix B. By multiplying $\mathbf{W}$ with both sides of eq. (10), we

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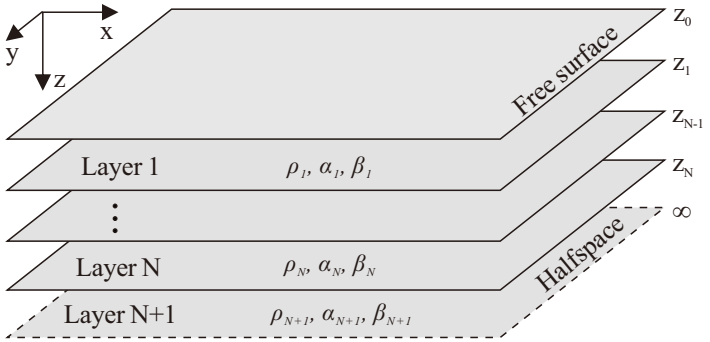


Figure 1. Illustration of the layered model. α, β and ρ denote the *P*-wave velocity, *S*-wave velocity and density.

obtain:

$$\omega^2\mathbf{W}\frac{\partial\mathbf{M}}{\partial m}\mathbf{U}-\mathbf{W}\frac{\partial\mathbf{E}}{\partial m}\mathbf{U}+\left(\omega^2\mathbf{WM}-\mathbf{WE}\right)\frac{\partial\mathbf{U}}{\partial m}$$
$$=2\gamma_{N+1}\frac{\partial\gamma_{N+1}}{\partial m}\mathbf{WK}_2\mathbf{U}+\gamma_{N+1}^2\mathbf{W}\frac{\partial\mathbf{K}_2}{\partial m}\mathbf{U}$$
$$+\frac{\partial\gamma_{N+1}}{\partial m}\mathbf{WK}_1\mathbf{U}+\gamma_{N+1}\mathbf{W}\frac{\partial\mathbf{K}_1}{\partial m}\mathbf{U}$$
$$+\left(\gamma_{N+1}^2\mathbf{WK}_2+\gamma_{N+1}\mathbf{WK}_1\right)\frac{\partial\mathbf{U}}{\partial m}. \tag{12}$$

Based on eq. (11), eq. (12) can be simplified as:

$$\omega^2\mathbf{W}\frac{\partial\mathbf{M}}{\partial m}\mathbf{U}-\mathbf{W}\frac{\partial\mathbf{E}}{\partial m}\mathbf{U}=2\gamma_{N+1}\frac{\partial\gamma_{N+1}}{\partial m}\mathbf{WK}_2\mathbf{U}+\gamma_{N+1}^2\mathbf{W}\frac{\partial\mathbf{K}_2}{\partial m}\mathbf{U}$$
$$+\frac{\partial\gamma_{N+1}}{\partial m}\mathbf{WK}_1\mathbf{U}+\gamma_{N+1}\mathbf{W}\frac{\partial\mathbf{K}_1}{\partial m}\mathbf{U}. \tag{13}$$

Then the derivative of γ_{N+1} can be expressed as:

$$\frac{\partial\gamma_{N+1}}{\partial m}=\frac{1}{\mathbf{W}\left(2\gamma_{N+1}\mathbf{K}_2+\mathbf{K}_1\right)\mathbf{U}}\mathbf{W}$$
$$\times\left[\omega^2\frac{\partial\mathbf{M}}{\partial m}-\left(\gamma_{N+1}^2\frac{\partial\mathbf{K}_2}{\partial m}+\gamma_{N+1}\frac{\partial\mathbf{K}_1}{\partial m}+\frac{\partial\mathbf{E}}{\partial m}\right)\right]\mathbf{U}. \tag{14}$$

The form of eq. (14) is a bit similar to the expressions of Haney & Tsai (2017), which were developed for the sensitivity analysis of normal modes. One of the main features of our method is the usage of the left eigenvectors of the eigenvalue system because the coefficient matrices ($\mathbf{M}$, $\mathbf{E}$, $\mathbf{K}_1$ and $\mathbf{K}_2$) are no longer symmetric. Once the partial derivative of the half-space vertical wavenumber γ_{N+1} is obtained, the partial derivative of the horizontal wavenumber k can be further calculated via the chain rule. Considering the wavenumber relation $k^2=\omega^2/\beta_{N+1}^2-\gamma_{N+1}^2$, we can obtain the partial derivative of k with

$$\frac{\partial k}{\partial m}=-\left(\omega^2\beta_{N+1}^{-3}\frac{\partial\beta_{N+1}}{\partial m}+\gamma_{N+1}\frac{\partial\gamma_{N+1}}{\partial m}\right)\Big/k. \tag{15}$$

For each model parameter m except the half-space shear-wave velocity, $\partial\beta_{N+1}/\partial m$ remains zero. Otherwise, $\partial\beta_{N+1}/\partial m=1$ and eq. (15) becomes:

$$\left.\frac{\partial k}{\partial m}\right|_{m=\beta_{N+1}}=-\left(\omega^2\beta_{N+1}^{-3}+\gamma_{N+1}\frac{\partial\gamma_{N+1}}{\partial m}\right)\Big/k. \tag{16}$$

Finally, based on the relation $v_c=\omega/\mathrm{Re}(k)$, the partial derivative of v_c can be obtained with

$$\frac{\partial v_c}{\partial m}=\frac{-\omega}{\mathrm{Re}(k)^2}\frac{\mathrm{Re}\left(\partial k\right)}{\partial m}, \tag{17}$$

When assuming m to be real values, eq. (17) can be further written as

$$\frac{\partial v_c}{\partial m}=\frac{-\omega}{\mathrm{Re}(k)^2}\mathrm{Re}\left(\frac{\partial k}{\partial m}\right). \tag{18}$$

With the above derivations, we can calculate the partial derivatives for all the normal and leaky modes solved by SASEM. Due to the usage of the intermediate results and coefficient matrices obtained during forward modelling, the sensitivity calculation has high accuracy and reliability identical to the forward modelling procedure.

2.2.2 *The sensitivity kernels*

Following Pan *et al.* (2019), the partial derivative of dispersion curves to the model parameters of a given layer, that is, $S_{ij}=\partial v_c^{(i)}/\partial m_j$ where $v_c^{(i)}$ denotes the phase velocity of the i-th mode and m_j denotes the parameters of the j-th layer, is called the phase-velocity sensitivity function. Meanwhile, we can define the sensitivity kernels $s_i(z)$ to reflect the depth-dependent sensitivities varying from the surface to the depth. The relation between the sensitivity kernel and the sensitivity function of a given layer is expressed as:

$$S_{ij}=\partial v_c^{(i)}\big/\partial m_j=\int_{z_{j-1}}^{z_j}s_i\left(z\right)\mathrm{d}z. \tag{19}$$

Numerically, the sensitivity kernel equals the sensitivity function of a thin layer with a unit thickness at a given depth. For the model composed of several homogenous layers, the sensitivity kernels are discontinuous at interfaces because of the variation of media parameters. It is worth noting that the SASEM will divide each layer into several elements that are controlled by Gauss–Lobatto–Legendre nodes. Because the *P*/*S* wave velocity and density of each node are independent variables, the SASEM can naturally calculate the partial derivatives to model parameters at different depths (nodes), that is, the discrete sensitivity kernel. The sensitivity function is then obtained by integral (weighted summation in codes) inside each layer. Compared to the traditional sensitivity analysis method based on the differentiation of the dispersion function (Cochran *et al.* 1970; Xia *et al.* 1999), SASEM can consider both the whole sensitivity and the sensitivity variation at different nodes within a given layer. From this perspective, SASEM has similar advantages to the analytical sensitivity analysis method based on the variational method (Aki & Richards 2002), as it can exhibit the detailed variation of the sensitivity at different depths without using dense layers.

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Because the magnitudes of different model parameters greatly influence the values of the corresponding sensitivity kernels (and functions), we further introduce the normalized sensitivity kernel (NSK; Kugler *et al.* 2007) which removes the effects of different parameter magnitudes and consequently shows the relative sensitivity of dispersion curves to model parameters. Typically, the NSK is defined as:

$$\bar{s}_i(z) = s_i(z) \cdot \frac{m(z)}{v_c^{(i)}}. \tag{20}$$

3 MODAL CLASSIFICATION

3.1 Guided-*P* modes and apparent Σ modes

In both seismic exploration and earthquake investigations, the dispersive waves arriving before surface waves have been noted (e.g. Oliver & Major 1960; Robertsson *et al.* 1996). Although often treated as acoustic waves in seismic exploration, the widely discussed 'guided waves', which propagate faster than surface waves and exhibit dispersion similar to acoustic waves, primarily belong to leaky waves in elastic wave theories (Roth *et al.* 1998; Sun *et al.* 2021). In this study, these acoustic-like waves are called guided-*P* waves because they are dominated by the *P* waves mainly guided by model interfaces, and their energy leakage is only related to converted *S* waves. In addition to the guided-*P* waves, another kind of dispersive waves often appear at lower frequencies (e.g. Gao *et al.* 2014; Lin *et al.* 2023), and depend more on *S* waves (Li *et al.* 2021).

To explain the different behaviours of leaky-wave dispersion, the features of leaky modes have long been studied (e.g. Cochran *et al.* 1970; Watson 1972; Wu *et al.* 2025). As Cochran *et al.* (1970) concluded, the leaky modes located on the (+, −) Riemann sheets exhibit complicated dispersion curves and can be classified into Π modes and Σ modes. Π modes have low attenuation and are sensitive to *P*-wave velocities. Σ modes have high attenuation and are sensitive to *S*-wave velocities. This classification method respectively gives modes sensitive to *P*- and *S*-wave velocities and naturally lends itself to inverse problems. With the Π/Σ classification, Cochran *et al.* (1970) discussed the dispersion features of a simple model whose velocity ranges of *P* and *S* waves do not overlap. For such models, Π modes are the only leaky-mode sets that primarily depend on *P*-wave velocities and can explain the guided-*P* waves well. However, the original Π/Σ classification exhibits some limitations for other models (see section 5.2). For the models whose velocity ranges of *P* and *S* waves overlap, Π modes may vanish at a certain frequency (i.e. the cut-off frequency) and convert to normal modes, which will further lead to the complicated mode-kissing phenomenon (Sun *et al.* 2021; Kennett 2023). Some modes located in the mode-kissing area contribute to the guided-*P* waves as well although they belong to normal modes. In other words, Π modes are incomplete for explaining guided-*P* waves. Consequently, a new mode category, named guided-*P* mode, is suggested. Guided-*P* modes are composed of Π leaky modes and the normal modes sensitive to *P*-wave velocities. With this definition, guided-*P* modes are naturally aligned with the purpose of *P*-wave velocity inversion. In some cases, the guided-*P* modes can be simply characterized using the acoustic normal modes. Acoustic normal modes are obtained by considering purely acoustic models whose *P*-wave velocities are identical to the investigated elastic models (Roth *et al.* 1998). For the models with very high Poisson's ratios, guided-*P* modes are almost identical to the acoustic normal modes. Remarkable differences between guided-*P* modes and acoustic normal modes are visible when the model structures are complex (e.g. with high or low-velocity layers) or Poisson's ratios are low. Compared with acoustic normal modes, the guided-*P* modes obtained with elastic theories can characterize the guided-*P* waves more accurately.

Although the Σ modes are not as attractive as guided-*P* modes, two subsets of Σ modes may be observable and worth attention. The most impressive Σ modes appear in the low-frequency range, which can exhibit ultra-low attenuation (García-Jerez & Sánchez-Sesma 2015) and explain the low-frequency and high-velocity dispersion energy in the dispersion spectra (Li *et al.* 2021; Lin *et al.* 2023). Besides, within a higher frequency range, a few Σ modes, whose phase velocities are slightly larger than the normal modes, exhibit relatively lower attenuation and have the potential to induce observable wave trains (Gao *et al.* 2014). These Σ modes that are discernible in the dispersion spectra are called the apparent Σ modes. But it should be noted that which Σ modes are apparent is related to the source and receiver parameters.

3.2 Modal classification algorithm

In leaky-mode observation, the most attractive modes are guided-*P* modes due to their ability to constrain *P*-wave velocities. When performing the widely used inversion schemes that match dispersion curves based on gradients, measuring the residuals between the theoretical guided-*P* modes and the corresponding observed ones is essential. Due to the strong attenuation, most Σ modes are invisible in field seismograms. Consequently, identifying and keeping guided-*P* modes can greatly simplify the match between the observed and theoretical dispersion curves. For simple models with high Poisson's ratios, guided-*P* and Σ modes can be distinguished empirically according to their attenuation coefficients (Shi *et al.* 2023; Wang *et al.* 2024). However, due to the lack of numerical criteria to distinguish different modes, it is challenging to perform modal classification for complicated dispersion curves whose attenuation coefficients of guided-*P* and Σ modes intersect or differ a little. Therefore, we are going to suggest a robust and effective strategy for the modal classification problem. The guided-*P* modes are first identified and the others are then automatically classified as Σ modes. We introduce two criteria to determine whether a mode belongs to guided-*P* modes.

(1) Amplitude ratio of *P*- and *S*-wave potentials

Among the complete leaky-mode solution, the guided-*P* modes are primarily composed of *P* waves whereas the other modes are primarily composed of *S* waves. The first criterion to distinguish guided-*P* modes is the amplitude ratio between the *P*- and *S*-wave potentials of each mode. Mathematically, the *P*- and *S*-wave potential functions of guided-*P* modes should satisfy:

$$\max|p(z)| > \max|q(z)| \qquad z \in [z_0, z_N], \tag{21}$$

where the potential wavefields can be determined using eqs (5) and (6).

(2) Normalized sensitivity kernels

Another criterion is based on the sensitivity kernels. As previous studies reported (Cochran *et al.* 1970; Li *et al.* 2021), guided-*P* modes are more sensitive to *P*-wave velocities. However, the raw sensitivity kernels present a challenge for mode identification, as the

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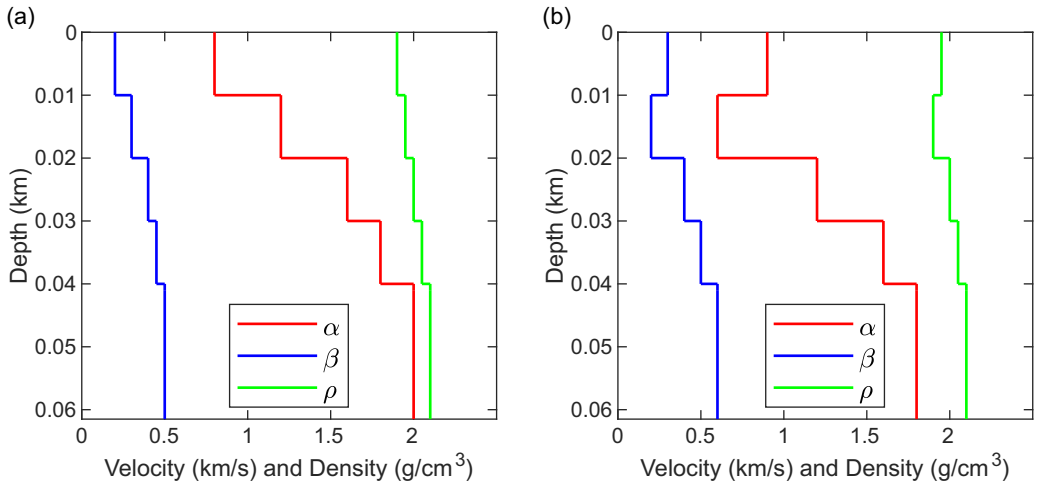


Figure 2. Structures of the (a) regular and (b) low-velocity-layer models.

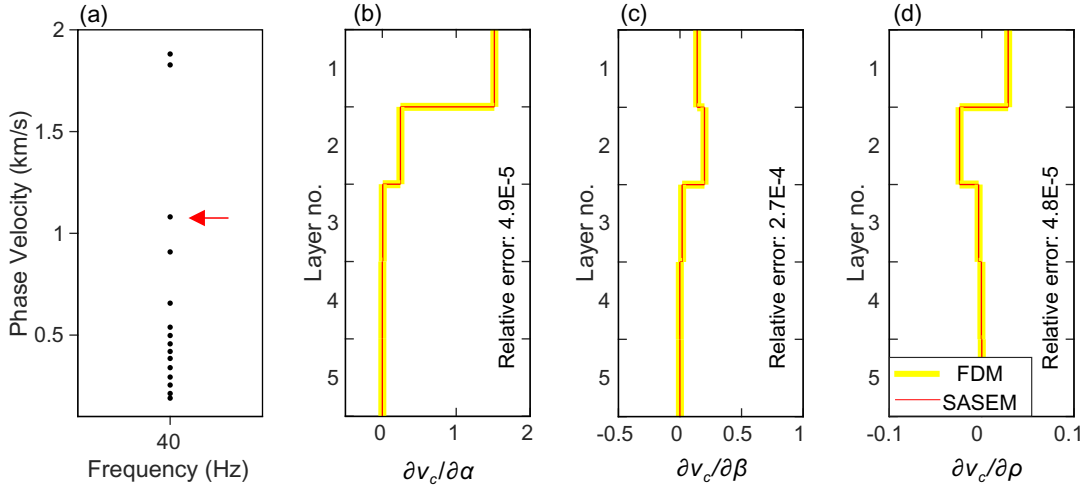


Figure 3. Numerical verification of the sensitivity calculation. (a) Modal solutions of the regular model at 40 Hz. The sensitivity functions of the mode marked by the arrow are calculated for comparison. (b)–(d) Comparison of the sensitivity functions calculated by SASEM and the finite-difference method (FDM).

magnitudes associated with different parameters vary significantly. This often results in the sensitivity kernel to *S*-wave velocities being much greater than those to *P*-wave velocities. Consequently, the normalized sensitivity kernel (NSK) defined with eq. (20) is employed. Assuming the i-th mode belongs to guided-*P* modes, its NSK should satisfy:

$$\max\left[\tilde{s}_i^{(\alpha)}(z)\right] > \eta \cdot \max\left[\tilde{s}_i^{(\beta)}(z)\right] \quad z \in [z_0, z_N], \tag{22}$$

where η denotes a constant usually set to 1. A smaller value (e.g. 0.8) is also an alternative if more modes with comparable sensitivity to *P*- and *S*-wave velocities are required. The normal and leaky modes that simultaneously satisfy the above criteria including eqs (21) and (22) are classified as guided-*P* modes.

3.3 Order identification of guided-*P* modes

According to the optimization algorithms used, dispersion inversion can usually be divided into two categories: local linearization inversion and nonlinear inversion. The local linearization inversion is preferred for retrieving complex 2-D/3-D models because of its higher computation efficiency. In surface-wave dispersion inversion based on local linearization, the order indexes of the observed and theoretical modes are required. Only with the correct order indexes, the misfit between the observed and theoretical dispersion curves can be correctly measured (Zhang & Chan 2003). For the observed modes, order identification is empirical and should be performed during extraction. However, the order identification for the theoretical modes has to be performed in each iteration after the dispersion curves are calculated. Based on the mode-solving process that can avoid root skipping, the order identification for normal modes is usually effortless because their order indexes can be obtained straightforwardly

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by sorting the phase velocities. However, such a method may not be effective for the identified guided-*P* modes due to the common mode skipping caused by modal classification (Section 4.2), particularly in models with low Poisson's ratios. Therefore, we propose an effective method to determine the order indexes of the calculated guided-*P* modes.

When solving modes with eq. (7), both the eigenvalues (wavenumbers) and the eigenvectors (wavefields) are calculated. The combination of the eigen wavenumber and wavefield can describe the propagation and vibration features of each mode. Numerically, the differences between the eigen wavefields of adjacent modes are more detectable compared to the eigen wavenumbers (or phase velocities). For example, when the well-known mode-kissing phenomenon occurs, the differences between the kissing modes are quite small, but differences between their eigen wavefields can be much more pronounced (Gao *et al.* 2016). Therefore, we determine the order of each guided-*P* mode according to their eigen wavefields. Given that the guided-*P* modes are predominantly composed of *P* waves, the eigen wavefields (*P*-wave potentials) of the acoustic normal modes can serve as reliable benchmarks for the order identification of guided-*P* modes. Similarly to PSV normal modes, the orders of acoustic normal modes can be naturally obtained by sorting their phase velocities or eigen wavenumbers. The main steps of order identification for a given guided-*P* mode are described as follows:

(i) *Solve (or import) the acoustic normal modes as well as the corresponding P-wave eigen potentials.*

(ii) *Calculate the normalized zero-lag cross-correlation functions of the given guided-P mode and the imported acoustic normal modes with:*

$$\mathrm{corr} = \frac{\left|\int_{z_0}^{z_N} p_g(z)\cdot p_a(z)\,\mathrm{d}z\right|}{\sqrt{\int_{z_0}^{z_N}|p_g(z)|^2\,\mathrm{d}z}\cdot\sqrt{\int_{z_0}^{z_N}|p_a(z)|^2\,\mathrm{d}z}}, \tag{23}$$

where $p_g(z)$ denotes the P-wave potential of the guided-P mode, and $p_a(z)$ denotes the potential of the acoustic normal mode.

(iii) *Find the acoustic normal mode with the largest cross-correlation function, and return the corresponding order index.*

It is worth noting that calculating and storing acoustic normal modes prior to solving leaky modes is advisable to prevent redundant calculations. Compared with the time cost for solving leaky modes, the time cost for solving acoustic normal modes is almost negligible (less than 10 per cent). Another issue worth attention is the possibility of identifying two adjacent guided-*P* modes in the same order. This can be addressed by reserving the mode with the highest sensitivity to *P*-wave velocities.

4 METHOD VERIFICATION

In this section, we perform numerical tests with different models to verify the effectiveness of the proposed methods including sensitivity calculation and modal classification. In addition to a regular model (the wave velocities increase with depth) which is commonly used in dispersion analysis, we also examine an irregular model with a low-velocity layer (LVL). Compared with the regular model, the dispersion curves and sensitivity features become complicated when the model contains a low-velocity layer.

4.1 Sensitivity calculation

First, the effectiveness of the proposed method is verified using the regular model in Fig. 2(a). For this numerical test, the reference values of the sensitivity functions are calculated using a finite-difference method:

$$S_{ij}^{(\mathrm{ref})} = \frac{v_c^{(i)}\left(m_j + \Delta m\right) - v_c^{(i)}\left(m_j\right)}{\Delta m}, \tag{24}$$

where Δm is a small value (0.01 per cent of m), and the phase velocity v_c is obtained by finding the zeros of the dispersion function with a high-precision root-searching algorithm (Wu & Chen 2017).

For the regular model, the normal- and leaky-mode solutions at 40 Hz are shown in Fig. 3(a). For the leaky mode marked by the arrow, the reference sensitivity functions and the results of the proposed method are compared in Figs 3(b)–(d), which demonstrates the correctness of the proposed method. It is noteworthy that the accuracy of the sensitivity depends primarily on the accuracy of the modal solutions determined by SASEM. In this test, there are seven nodes in each element, which has been proven to be able to achieve high-precision modal solutions (Shi *et al.* 2022).

4.2 Modal classification and order identification

For the regular model in Fig. 2(a), we calculate the normal- and leaky-mode dispersion curves in the frequency range of 1–100 Hz with SASEM (shown in Fig. 4a). Fig. 4(b) shows the NSKs of the modes at 40 Hz which are composed of 9 normal modes and 6 leaky modes. All the normal modes (mode indexes between 0 and 8 in Fig. 4b) are more sensitive to shear-wave velocities, which is consistent with previous studies on surface-wave dispersion. The 6 leaky modes exhibit different sensitivity features: Modes 9 to 11 and Mode 14 are more sensitive to *S*-wave velocities; Modes 12 and 13 are more sensitive to *P*-wave velocities (Fig. 4b) and exhibit weak attenuation (Fig. 4a). Meanwhile, the *P*-wave potentials of Modes 12 and 13 have larger amplitudes than the *S*-wave potentials (Fig. 4d). According to our former definitions, Modes 9 to 11 and Mode 14 belong to Σ modes while Modes 12 and 13 are classified as guided-*P* modes.

Fig. 4(c) shows the modal classification results obtained based on the proposed criteria. The guided-*P* modes and Σ modes have been successfully identified and shown with different markers in Fig. 4(c). In this model, the guided-*P* modes show quite different phase-velocity range from the normal modes and different attenuation-coefficient range from the Σ modes. Consequently, these guided-*P* modes are quite easy to identify. Moreover, the guided-*P* modes of different orders are marked with different colours. Besides the *P*- and *S*-wave potential functions of the 2 identified guided-*P* modes at 40 Hz, Fig. 4(d) shows the first 2 acoustic normal modes. Remarkably, the *P*-wave potential functions of guided-*P* modes are highly related to those of the acoustic normal modes.

The dispersion curves of the LVL model appear to be much more complicated than those of the regular model. Within the region of leaky modes, the interconversion between the guided-*P* modes and the Σ modes may occur, particularly in the high-frequency range, as highlighted by the box in Fig. 5(a). The interconversion and mode kissing leads to challenges for the experiential classification according to phase velocities and attenuation coefficients. The automatic modal classification method is used to treat this case. The NSKs of the 15 calculated modes at 40 Hz are shown in Fig. 5(b), where

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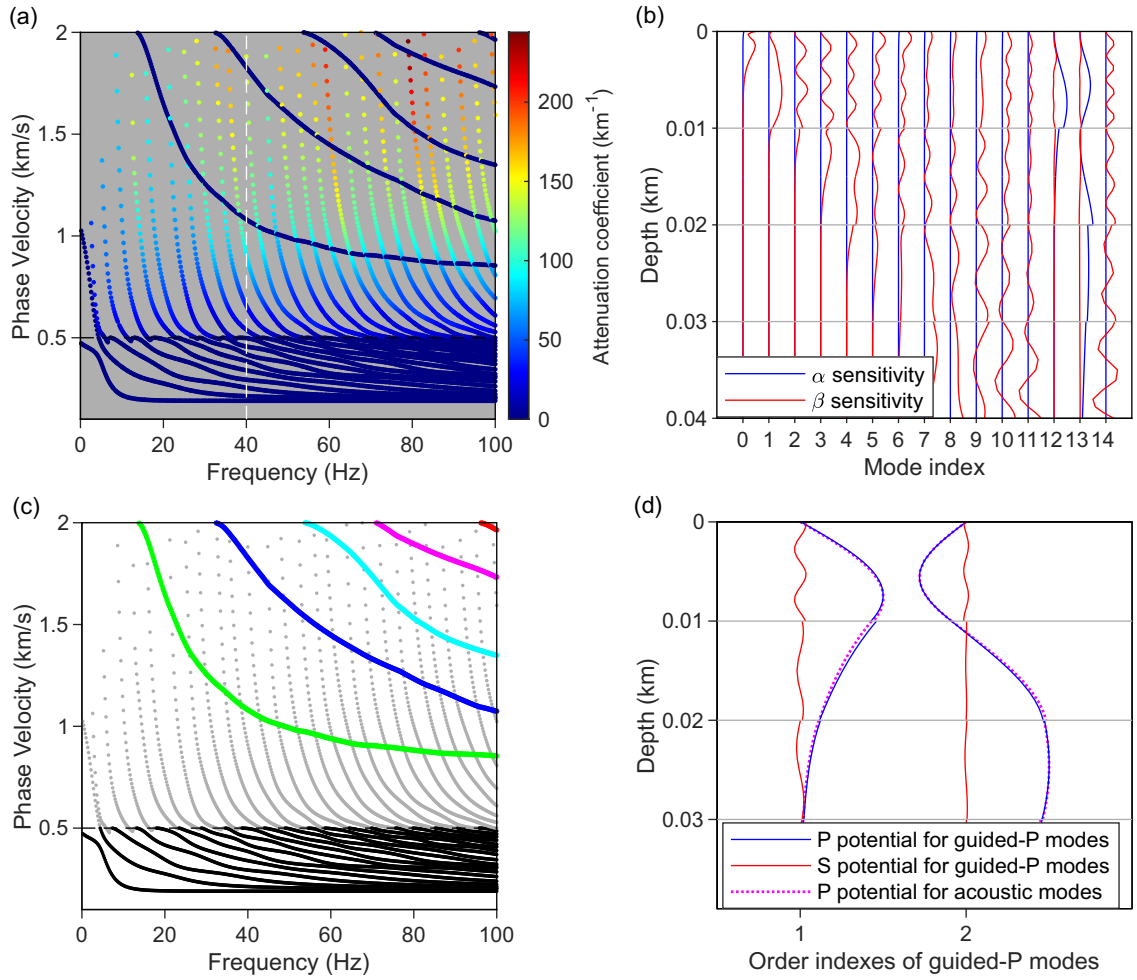


Figure 4. Modal classification for the regular model. (a) Modal solutions from 1 to 100 Hz with an interval of 0.2 Hz. (b) Normalized sensitivity kernels of the modes at 40 Hz. (c) Modal classification results. The black and grey dots denote normal and Σ modes respectively. The identified guided-P modes are denoted by coloured asterisks. The guided-*P* modes of the same order are marked by the same colour. (d) The *P*-wave potentials and *S*-wave potentials of the 2 guided-*P* modes at 40 Hz. The dashed lines denote the potentials of the acoustic normal modes (benchmark). The horizontal lines in panels (b) and (d) denote the interfaces of layers. Note that the NSKs and potentials are discontinuous when passing through the interfaces.

Modes 10, 12 and 14 exhibit higher sensitivity to *P*-wave velocities. Fig. 5(c) shows the modal classification results obtained based on the proposed criteria. In the frequency range from 1 to 100 Hz, 7 dispersion curves of guided-*P* modes are identified. Fig. 5(d) shows the *P*-wave potential functions of the 3 identified guided-*P* modes at 40 Hz. For guided-*P* modes, the *P*-wave potentials are highly related to those of the acoustic normal modes as well. Notably, dispersion curves of the identified guided-*P* modes can be discontinuous (see Fig. 5e). Consequently, simply sorting the phase velocities leads to failures in determining the order indexes of guided-*P* modes (see Fig. 5f).

To further prove the practicality of the proposed mode classification method, we compare the identified guided-*P* modes with synthetic dispersion spectra. Following Wang *et al.* (2019) and Li *et al.* (2021), the theoretical dispersion spectrum (TDS) which is composed of the kernel functions of the generalized reflection and transmission method (Chen 1999) is used as the benchmark. For the regular and LVL models, we synthesized their TDS with idealized explosion sources placed at a depth of 5 m and receiver arrays arranged on the surface. For the regular model, the TDS and the dispersion curves of the multiple guided-*P* modes are shown in Fig. 6(a). Because of the simplicity of the regular model, the dispersion energy of different modes is quite clear and smooth. The identified guided-*P* modes are highly consistent with the dispersion energy. As well, the identified guided-*P* dispersion curves can perfectly match the dispersion energy although the TDS for the LVL model (Fig. 6b) exhibits complicated dispersion features, which indicates that the proposed method for modal classification is effective in providing reliable guided-*P* dispersion curves. Besides the remarkable guided-*P* mode dispersion, the TDS of the LVL model also exhibits the dispersion energy of the apparent Σ mode at the low-frequency range. As shown in Fig. 6(b), the low-frequency dispersion energy intersects with the dashed line representing the half-space *S*-wave velocity, which can be explained well with the 1st normal mode and the connected apparent Σ mode (see Appendix C).

5 NUMERICAL EXAMPLES FOR SENSITIVITY ANALYSES

In this section, the proposed method is applied to calculate the modal sensitivity for more practical models. Due to the frequent reports

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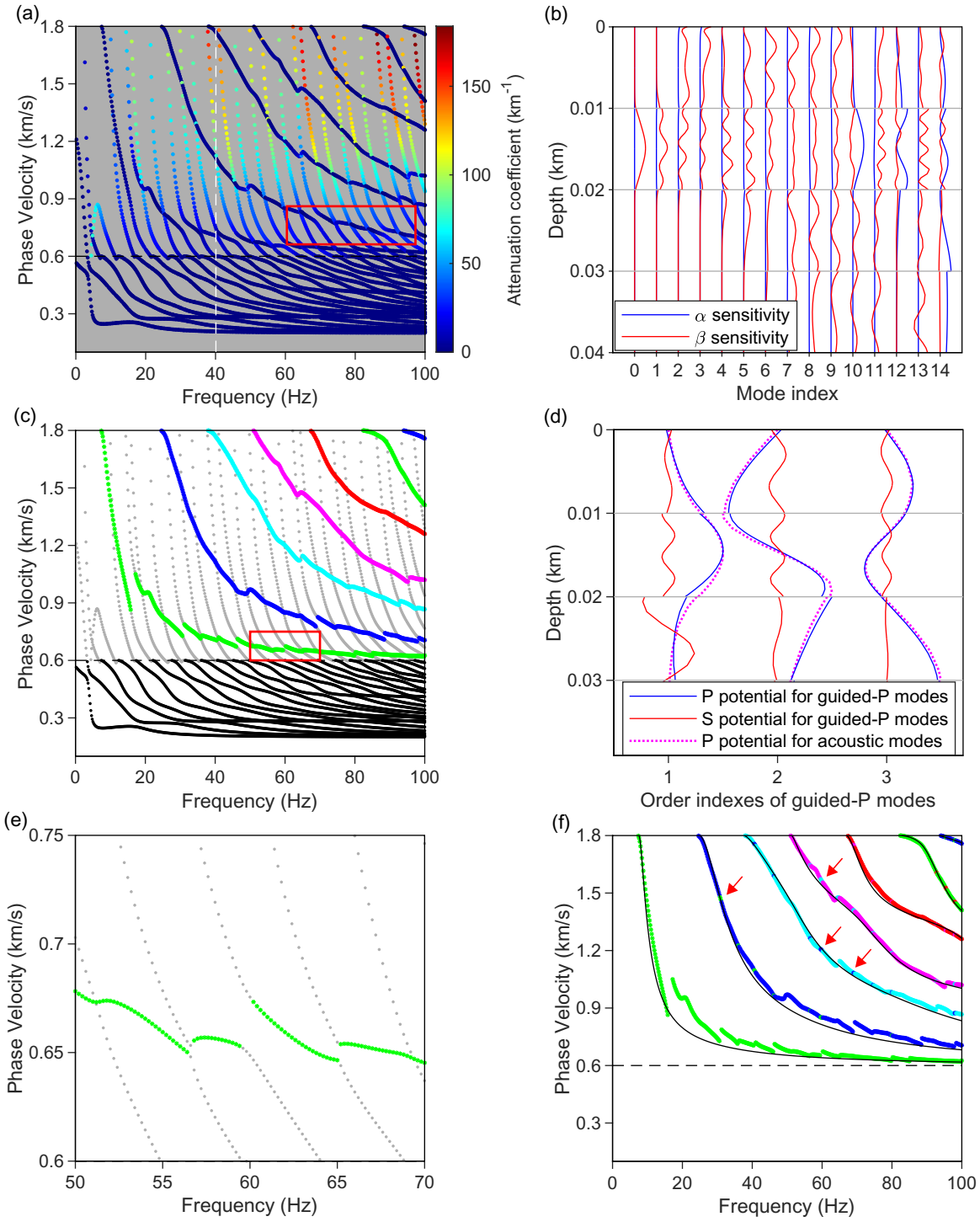


Figure 5. Modal classification for the LVL model. (a) Modal solutions from 1 to 100 Hz with an interval of 0.2 Hz. (b) Normalized sensitivity kernels of the modes at 40 Hz. (c) Modal classification results. Explanations of the markers and lines are identical to Fig. 4(c). (d) *P*-wave and *S*-wave potentials of the 3 guided-*P* modes at 40 Hz. The dashed lines denote the potentials of the acoustic normal modes. (e) Modes inside the box in panel (c). (f) Incorrect order indexes of guided-*P* modes obtained by sorting phase velocities. Several modes that are incorrectly ordered are marked by arrows. The solid lines denote the dispersion curves of acoustic normal modes.

and increasing interest in leaky modes from natural and exploration seismology in recent years, two synthetic models including a crust-scale model and a near-surface model are discussed, aiming to guide the inversion for different cases.

5.1 A crust and uppermost mantle model

Since the 1960s, observations of the weak leaky modes between the *P* and *S* waves in earthquake signals have been extensively documented (e.g. Oliver & Major 1960; Su & Dorman 1965; Ibrahim

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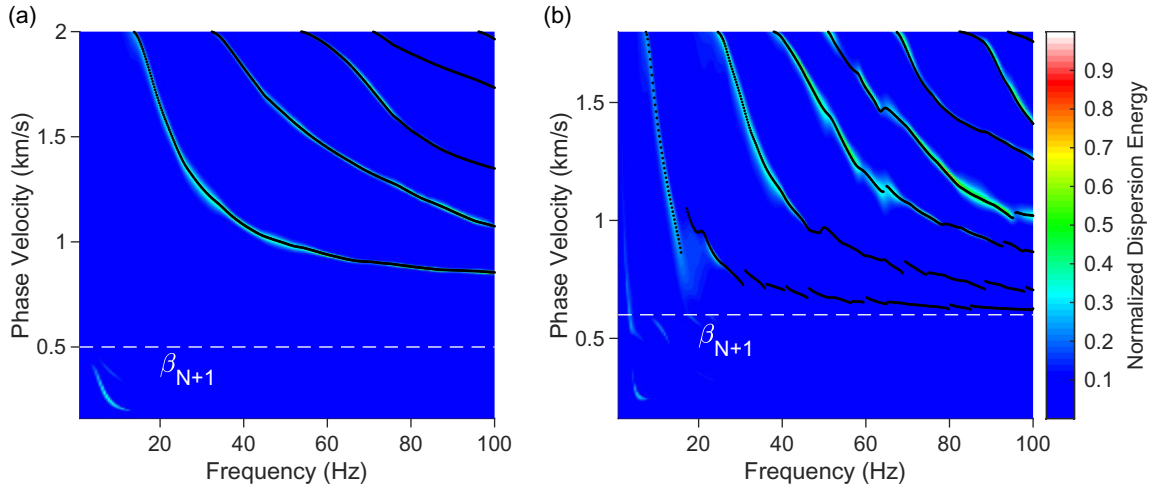


Figure 6. Theoretical dispersion spectra for the vertical displacement. (a) The regular model. (b) The low-velocity-layer model. Black dots denote the identified guided-*P* modes.

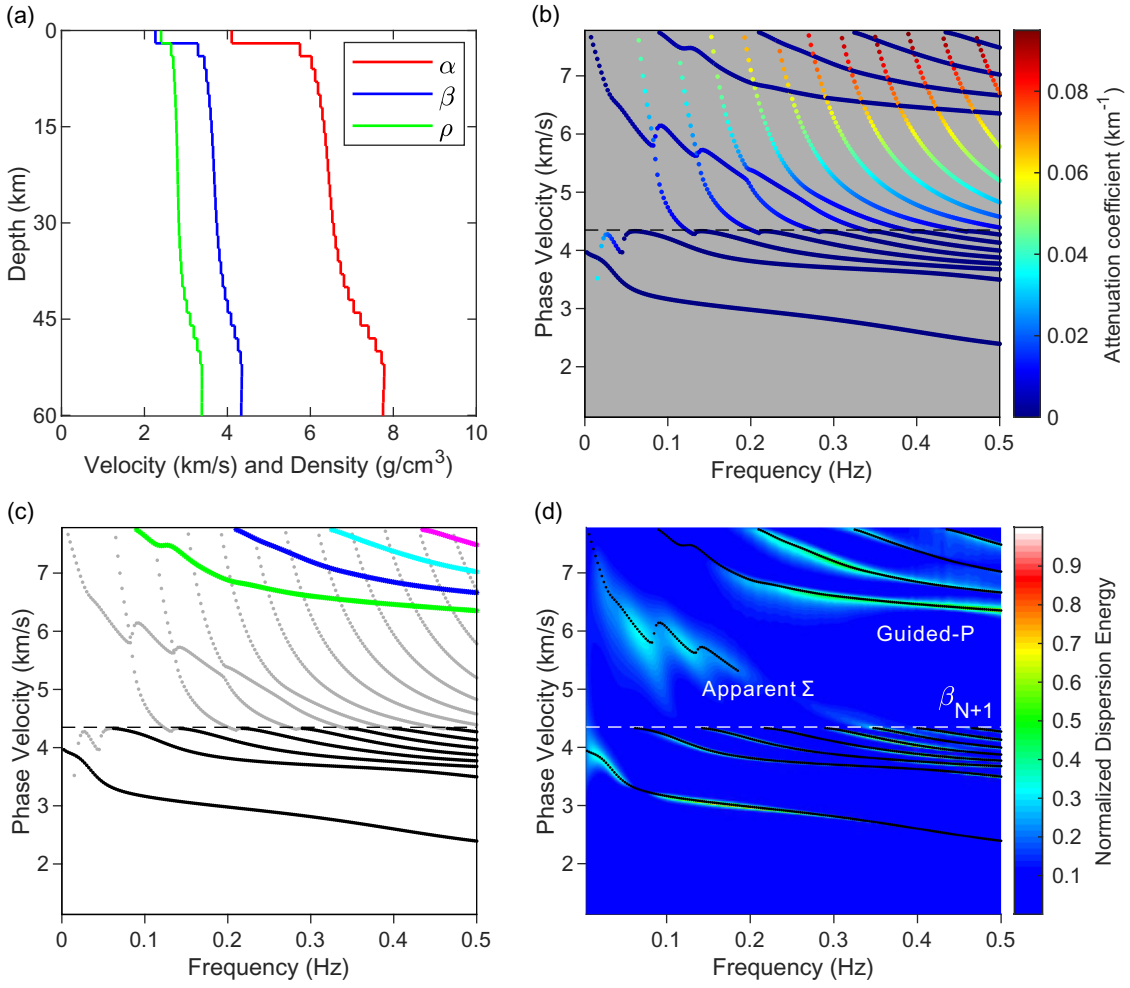


Figure 7. Structures and dispersion of the crust and uppermost mantle model. (a) Model parameters. (b) Modal solutions. (c) Modal classification results. Explanations of the markers are identical to Fig. 4(c). (d) Theoretical dispersion spectra for the vertical displacement.

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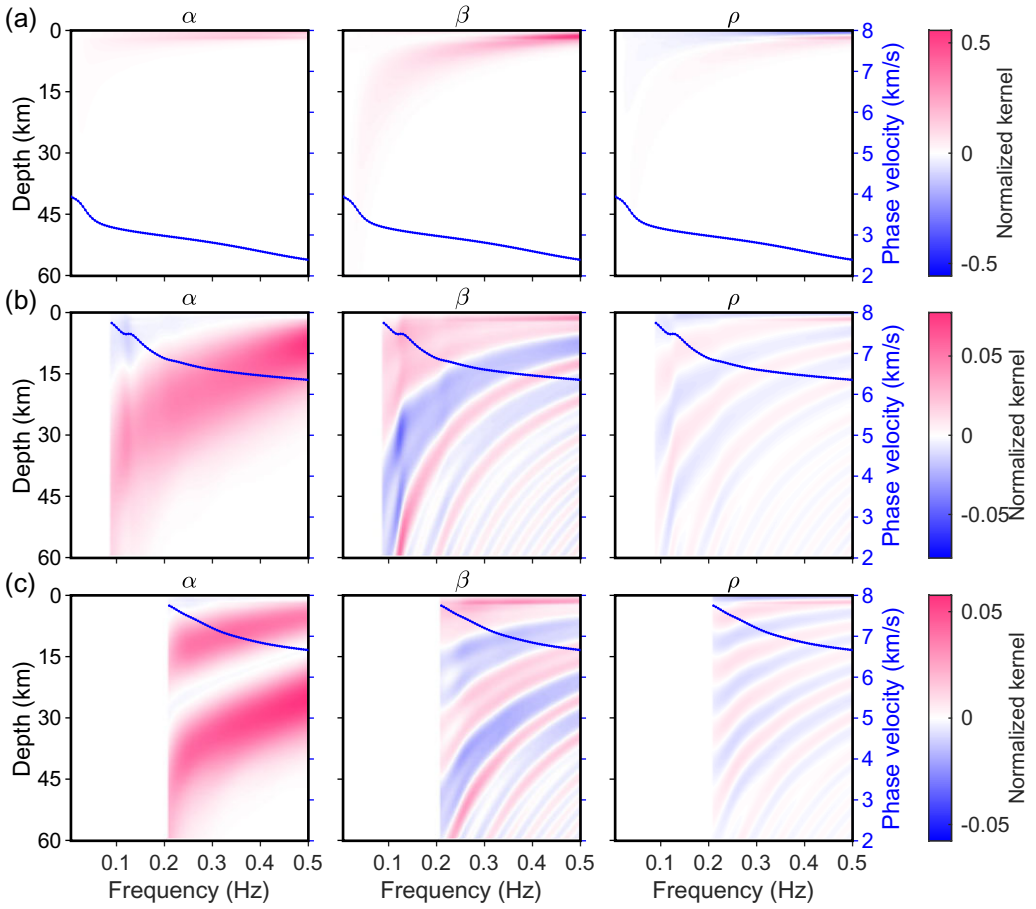


Figure 8. Normalized sensitivity kernels of the normal and guided-*P* modes for the crust and uppermost mantle model. (a) The fundamental normal mode. (b) The first guided-*P* mode. (c) The second guided-*P* mode.

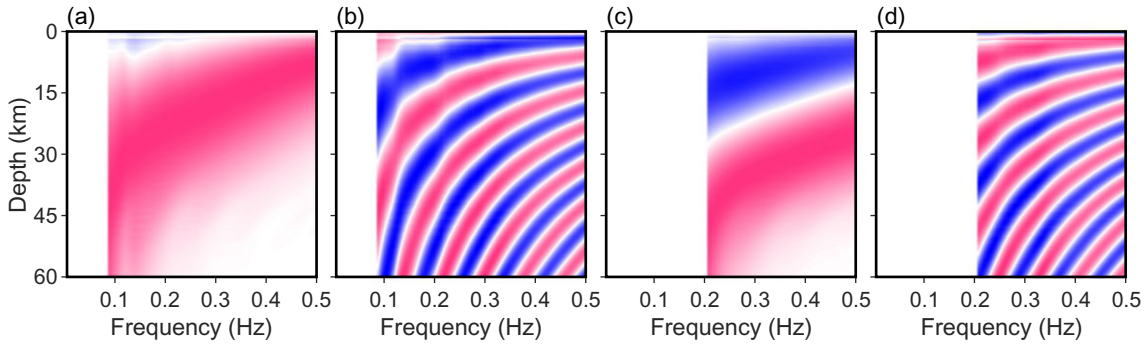


Figure 9. *P*- and *S*-wave potentials of guided-*P* modes. (a) and (b) are *P*- and *S*-wave potentials of the first guided-*P* mode. (c) and (d) are *P*- and *S*-wave potentials of the second guided-*P* mode. Both the *P*- and *S*-wave potentials have been normalized independently at each frequency.

1969; Fujita & Nishimura 1991). With a windowed Frequency–Bessel transform and the seismograms recorded by a dense array, Li *et al.* (2021) successfully extracted multiple leaky-mode dispersion and demonstrated their sensitivities to the underground structures. In this section, we study the leaky-mode sensitivities with a more accurate method.

Fig. 7(a) shows a crust and uppermost mantle model in the middle of the United States, which is modified from the results given by Shen & Ritzwoller (2016). The model is composed of 30 2-km-thick homogeneous layers and an underlying half-space layer. The first layer with the lowest velocities is used to simulate the shallow sediment. With SASEM, we obtain the modal solutions in the frequency range of 0–0.5 Hz (Fig. 7b). These modes, categorized as normal modes, guided-*P* modes, and Σ modes, are differentiated using various markers in Fig. 7(c). For real earthquakes, the excitation of leaky modes highly depends on the source depths and mechanisms (Wu *et al.* 2025). Typically, deep sources favour the excitation of leaky modes. In this study, a double-couple point source representing the earthquake mechanism whose strike, dip and rake angles are all 45° is placed at a depth of 10 km to study the

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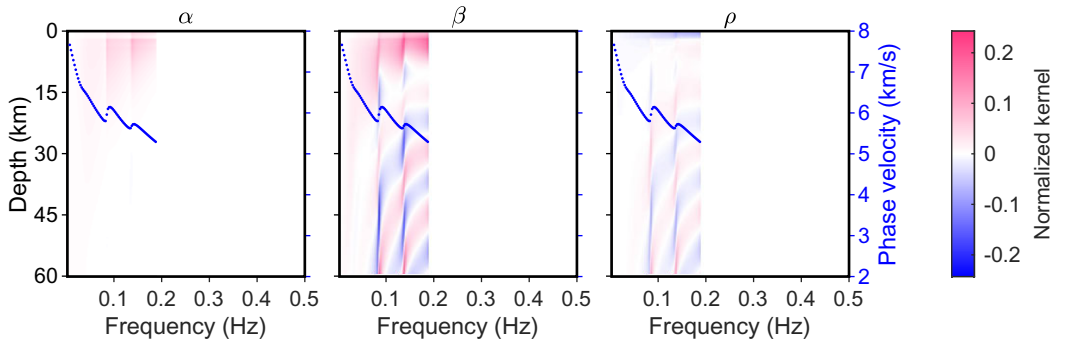


Figure 10. Normalized sensitivity kernels of the apparent Σ modes for the crust and uppermost mantle model.

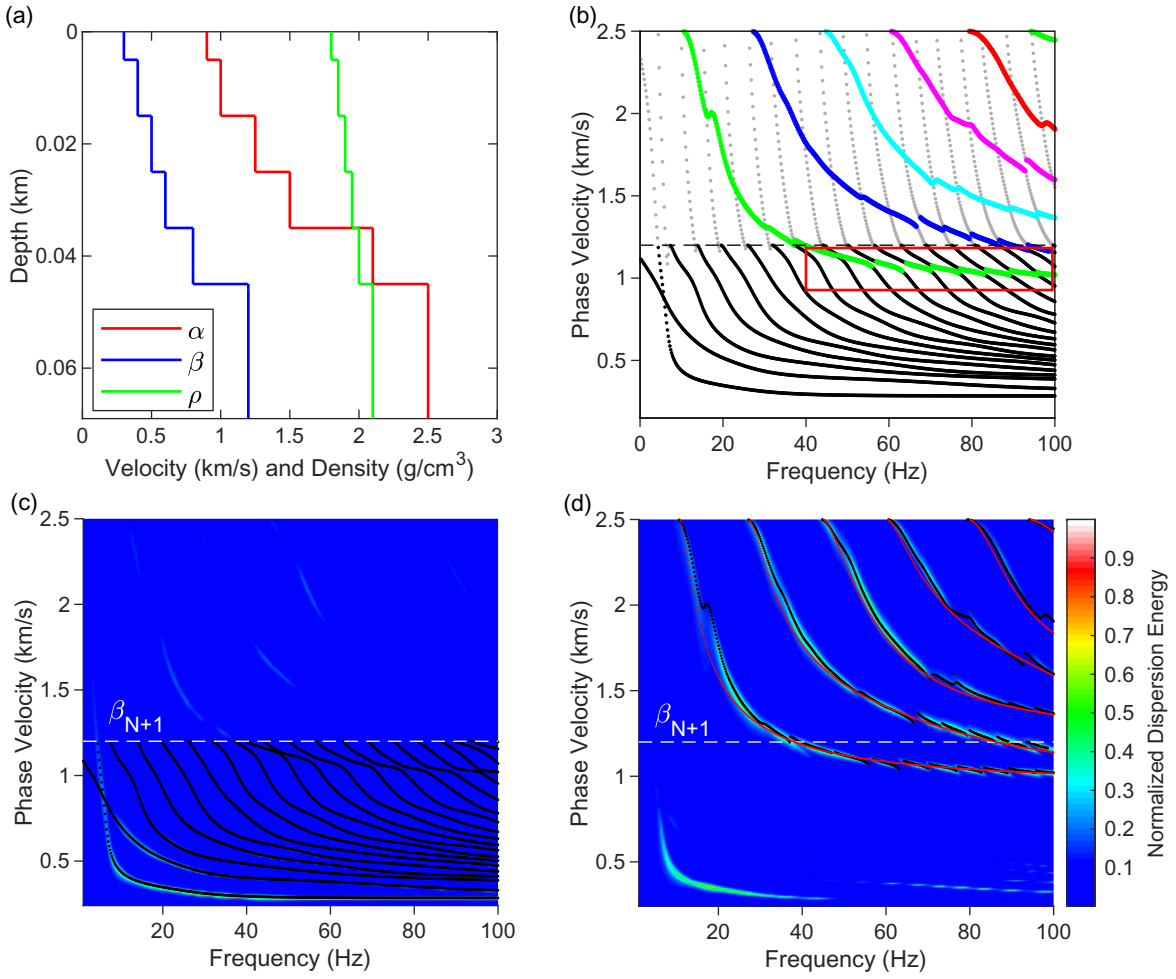


Figure 11. The structure and dispersion of the near-surface model. (a) Model parameters. (b) Modal solutions and classification results. Explanations of the markers are identical to Fig. 4(c). (c) Theoretical dispersion spectra for the vertical displacement using a vertical single-force source. Black dots represent the normal modes. (d) Theoretical dispersion spectra using an explosive source. Black and red dots denote the guided-*P* modes and acoustic normal modes respectively.

dispersion features of earthquake signals. The TDS shown in Fig. 7(d) once again validates the effectiveness of the modal classification for guided-*P* modes.

5.1.1 Sensitivity of normal and guided-P modes

For the fundamental normal mode and the two lowest guided-*P* modes, Fig. 8 shows the NSKs of the phase-velocity dispersion curves. As has been reported in many previous studies, the phase velocities of the fundamental mode are most sensitive to *S*-wave velocities (e.g. Xia *et al.* 2003; Pan *et al.* 2019). Figs 8(b) and (c) show the NSKs of the lowest two guided-*P* modes, which exhibit high sensitivity to *P*-wave velocities. Because the normal and leaky modes are both the solutions of the PSV-wave equations, they will be affected by the model parameters (including *P*/*S*-wave velocities and density) to varying degrees. Compared with the offshore

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model studied by Shi *et al.* (2023), the coupling between *P* and *S* waves in the crust and uppermost mantle model is much stronger. A notable feature is that the fundamental mode of this model is also sensitive to α to a certain degree, which may arise from the low velocity of the first layer that forms excellent waveguides for both *P* and *S* waves. Meanwhile, guided-*P* modes exhibit remarkable sensitivity to β, indicating that the guided-*P* modes depend on *S*-wave velocities as well. Consequently, oversimplifying the dispersion of guided-*P* (or normal) modes as functions solely dependent on *P* (or *S*) wave velocities in inversion may lead to significant inaccuracy.

Comparing the sensitivity of the first and second guided-*P* modes reveals that the higher guided-*P* modes exhibit more sensitivity to the *P*-wave velocities at deeper parts, which is foreseeable by analogy with surface waves. Besides, the sensitivity shows some features that are quite different from normal modes. It is well-known that normal modes are generally positively sensitive to the velocities, which means higher *P*- or *S*-wave velocities result in higher phase velocities (e.g. Pan *et al.* 2019). In contrast, guided-*P* modes may exhibit negative sensitivities to both the *P*- and *S*-wave velocities. While the sensitivity to α predominantly exhibits negative values only in a very small region (left-up corner of the first sensitivity image in Fig. 8b), clusters of negative values repeatedly appear in the sensitivity to β, which indicates quite complicated relations between β and the phase velocities of the guided-*P* modes. Additionally, because neither the normal modes nor the guided-*P* modes are primarily sensitive to density, it is still recommended to treat the density as a passively determined or empirical parameter in inversion.

For guided-*P* modes, an interesting feature is that the sensitivity to β has higher vertical resolutions and larger sensitive depth than the sensitivity to α. As the frequency increases, guided-*P* modes show weaker sensitivity to the *P*-wave velocities in deep parts, which is similar to the features of normal-mode dispersion. However, the guided-*P*-mode sensitivity to the deep *S*-wave velocities is always remarkable. The rationality of resolution and sensitive depth can be inferred via eq. (14) which describes the relations between the sensitivity kernel and the eigen wavefields. The eigen wavefields of the first and second guided-*P* modes, including the *P*- and *S*-wave potentials, are shown in Fig. 9. First, there is a notable relativity between the eigen wavefields vibration and the vertical resolution of sensitivity kernels. The high vertical resolutions for *S*-wave velocities are consistent with the rapid vibration of the *S*-wave potentials. Besides, the *S*-wave vibration of the guided-*P* modes and the sensitivity kernels to *S*-wave velocities can access the underlying half-space. From this point of view, constraining *S*-wave velocities with guided-*P* modes are similar to the surface-wave inversion with multiple higher normal modes to some degree, which may help to retrieve *S*-wave velocity structures with higher resolutions and larger inversion depth.

5.1.2 *Apparent* Σ *modes*

Besides the guided-*P* modes, the TDS in Fig. 7(d) shows extra leaky-mode energy in the frequency of 0–0.2 Hz. Based on the modal classification shown in Fig. 7(c), this leaky-mode energy is excited by the apparent Σ modes. Because of the strong attenuation, most of the Σ modes contribute little to wavefields and are usually invisible in dispersion spectra. However, the existence of the low-velocity sedimentary layer may favour the excitation of apparent Σ modes that exhibit remarkable dispersion energy at low frequencies. Combining the TDS, we extract the low-frequency apparent Σ modes from the leaky-mode solutions. Their sensitivity kernels are shown in Fig. 10, which indicates that the apparent Σ modes are sensitive to the *S*-wave velocities from shallow parts to depth. However, it is difficult to identify the order indexes of apparent Σ modes for both the observed modes and the theoretical modes, which greatly affects the application of local linearization inversion. Inversion workflows that are free of order identification (e.g. Yan *et al.* 2022; Liu *et al.* 2024) are alternatives. The nonlinear inversion algorithms based on dispersion functions (e.g. Maraschini & Foti, 2010) may also be effective for the inversion using apparent Σ modes, which, however, loses the advantage of low computation cost.

5.2 A near-surface model

As has been reported, the Poisson's ratios greatly influence the behaviours of the leaky modes (Sun *et al.* 2021). For the models with high Poisson's ratios (e.g. Fig. 2a), the guided-*P* mode dispersion is quite similar to purely acoustic modes (Roth & Holliger 1999). Meanwhile, different kinds of leaky modes can be identified quite easily according to their attenuation coefficients. As the Poisson's ratio decreases, the interaction of *P* and *S* waves influences the dispersion curves remarkably. Specifically, for the models whose *P*- and *S*-wave velocity ranges overlap, the *P*-wave-dominant modes and *S*-wave-dominant modes may interact and lead to complex mode-kissing phenomena (Kennett 2023). The dispersion and modal sensitivity for such models have rarely been reported due to the lack of observation in past years. Recently, several experiments on ice using fibre optic sensing and seismic nodes (e.g. Fichtner *et al.* 2023; Yang *et al.* 2024; Zhang *et al.* 2024) show interactions of surface- and leaky-wave dispersion. For better inversion, the dispersion and sensitivity analysis on such models is valuable.

Fig. 11(a) shows a model where there is a significant overlap between *P*- and *S*-wave velocities. With the proposed method, normal and leaky modes in the frequency range of 1–100 Hz are calculated and classified. Fig. 11(b) shows that the identified guided-*P* modes extend into the phase-velocity range of normal modes. For example, modes inside the box of Fig. 11(b), whose phase velocity is lower than the half-space shear-wave velocity β_{N+1}, theoretically belong to normal modes. However, some of these modes are identified as guided-*P* modes according to our classification criteria. Due to the interaction of *P*- and *S*-wave dispersion, the identified guided-*P* modes with phase velocities lower than β_{N+1} are composed of many short normal-mode branches (Kennett 2023). Nevertheless, the final obtained guided-*P* dispersion curves exhibit overall trends consistent with acoustic dispersion curves (Fig. 11d).

Because of the widespread usage in near-surface exploration, the hammer (vertical single force) and explosive sources were both employed to simulate the dispersion spectra. Fig. 11(c) shows the TDS of a vertical single-force source on the surface. In Fig. 11(c), the calculated normal modes agree well with the surface-wave dispersion energy. However, guided-*P* dispersion energy is quite weak because of the dominance of surface waves in the TDS. It has been reported that the guided-*P* mode energy becomes remarkable when using an explosive source, especially in offshore environments (e.g. Tran *et al.* 2024). Fig. 11(d) shows the TDS of an explosive source located at a depth of 5 m. It can be recognized that the dispersion energy in TDS agrees with the guided-*P* modes much better than acoustic normal modes. Furthermore, the dispersion energy

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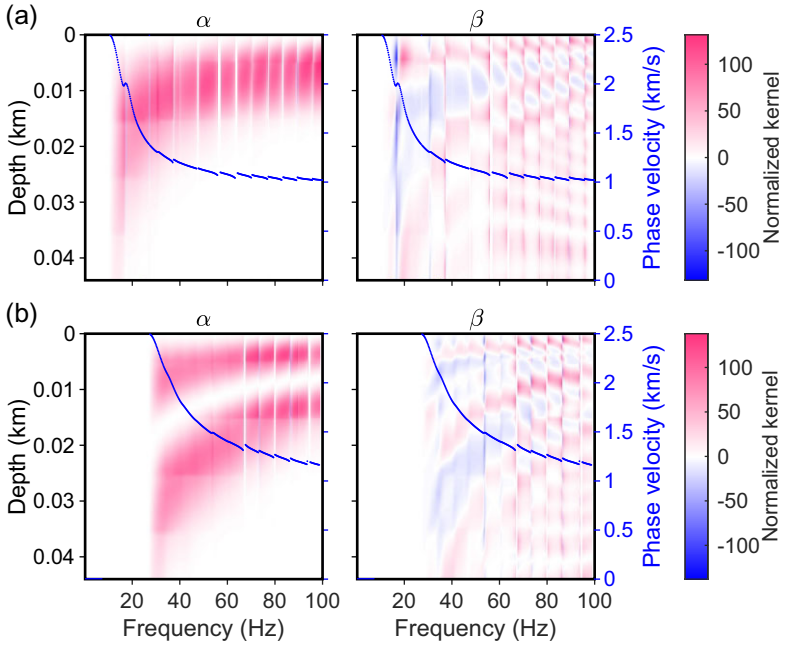


Figure 12. Normalized sensitivity kernels of the guided-*P* modes for the near-surface model. (a) The first guided-*P* mode. (b) The second guided-*P* mode.

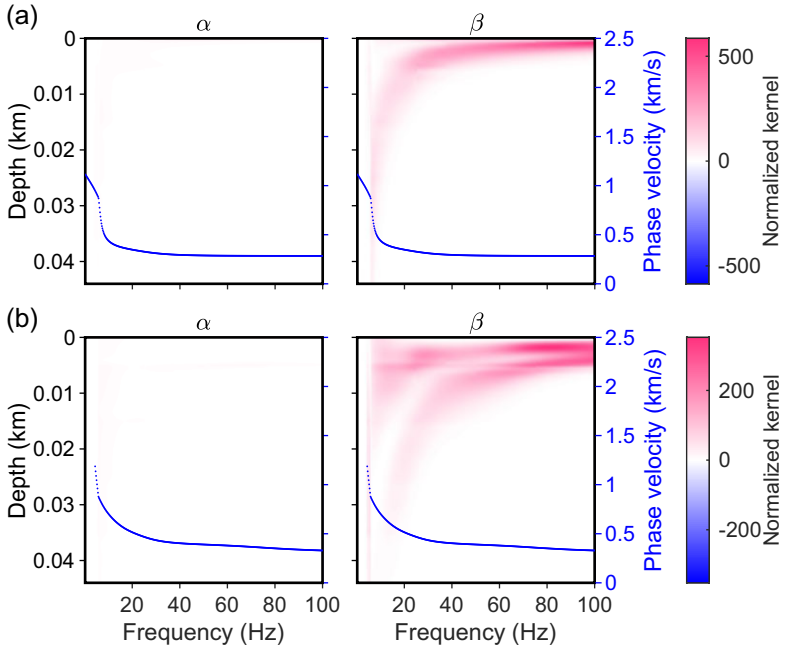


Figure 13. Normalized sensitivity kernels of the normal modes for the near-surface model. (a) The fundamental normal mode. (b) The first normal mode.

in the TDS exhibits discontinuity in the position identical to the theoretical guided-*P* modes, which again verifies the effectiveness of the strategy for identifying guided-*P* modes.

For all the modes shown in Fig. 11(b), the NSKs can be calculated accurately with the proposed method. Fig. 12 shows the NSKs of the first and second guided-*P* modes. Besides the strongest sensitivity to *P*-wave velocities, the guided-*P* modes are also affected by *S*-wave velocities. In contrast, the lower normal modes are almost insensitive to *P*-wave velocities (Fig. 13). In the NSKs of guided-*P* modes, there are many vertical white stripes that are related to the mode missing in dispersion curves. As a result, the sensitivity kernels of guided-*P* modes lose continuity and become more complicated. However, by ignoring these white stripes, the sensitivity images exhibit smooth variation with frequency, which indicates the rationality of the identified guided-*P* modes. Although we lose some sensitivity information at certain frequencies, the remaining sensitivity is distributed well at different depths. This means enough constraints to retrieve model structures if we have extracted abundant dispersion curves from field data. On this basis, the traditional dispersion inversion strategy using local optimization methods can still work well for such models.

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6 CONCLUSIONS

With the development of observation and dispersion detection methods, leaky modes have been reported frequently in recent years and applied to constrain the *P*-wave velocities (e.g. Li *et al.* 2022; Shi *et al.* 2023; Wang *et al.* 2024). The accurate calculation of the leaky-mode sensitivity kernels and the automatic classification of different kinds of modes are important for inversion. In this study, a novel and effective method for the sensitivity analysis of leaky modes has been developed. To calculate the sensitivity kernels accurately and stably, we derived concise formulae for the partial derivatives with the previously proposed semi-analytical spectral element method. Based on the sensitivity kernels, we suggest a modal classification strategy that can identify the attractive guided-*P* modes from the complicated normal- and leaky-mode solutions, which is one of the keys to performing inversion.

After the numerical verification using the classical regular and LVL models, we perform the sensitivity analysis for typical crust-scale and near-surface models. With the analysis of the crust and uppermost mantle model, we are aware that the guided-*P* modes cannot only constrain the crustal *P*-wave velocities well but also exhibit complemented and high-resolution sensitivities for *S*-wave velocities in deep parts, which indicates the necessity of joint inversion using full elastic theories. The sensitivity analysis for the near-surface model with overlapping *P*- and *S*-wave velocity ranges demonstrates the effectiveness of the proposed method for the complicated and discontinuous guided-*P* dispersion curves. With the stable sensitivity calculation and modal classification methods, subsequent joint inversion for complicated crust-scale and near-surface models can be performed with the existing inversion methods as described in Shi *et al.* (2023) and Wang *et al.* (2024).

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DATA AVAILABILITY

No field data is used in this paper. The Matlab codes to perform sensitivity analysis are available at https://doi.org/10.5281/zenodo.14286748.

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APPENDIX A: DERIVATION OF THE QUADRATIC EIGENVALUE PROBLEM

In the following derivations, we propose an improved method based on Shi *et al.* (2022), which will result in simpler eigenvalue problems and be friendly for further sensitivity analyses. First, the stratified model is divided into two regions: Region 2 contains the half-space substrate, and Region 1 contains the rest of the layers. The wavefields in Region 1 are described using the nodal displacements while the wavefields in Region 2 are described using *P*- and *S*-wave potentials (p and q). We multiply wavenumber k with the weak form of eqs (1) and (2) and let $\bar{U}_r{=}kU_r$, $\bar{q} = kq$. Then we obtain

$$\begin{aligned}&\int_{z_0}^{z_N} \rho\omega^2 \bar{U}_r \varphi \mathrm{d}z - k^2 \int_{z_0}^{z_N} (\lambda + 2\mu)\, \bar{U}_r \varphi \mathrm{d}z \\ &\quad -k^2 \int_{z_0}^{z_N} \left(\lambda \frac{\partial U_z}{\partial z}\varphi - \mu U_z \frac{\partial \varphi}{\partial z}\right) \mathrm{d}z \\ &\quad - \int_{z_0}^{z_N} \mu \frac{\partial \bar{U}_r}{\partial z}\frac{\partial \varphi}{\partial z}\mathrm{d}z \quad + [\tau_r \varphi]_{z_0}^{z_N} = 0, \end{aligned} \tag{A1}$$

$$\begin{aligned}&\int_{z_0}^{z_N} \rho\omega^2 U_z \varphi \mathrm{d}z - k^2 \int_{z_0}^{z_N} \mu U_z \varphi \mathrm{d}z + \int_{z_0}^{z_N} \left(\mu \frac{\partial \bar{U}_r}{\partial z}\varphi - \lambda \bar{U}_r \frac{\partial \varphi}{\partial z}\right) \mathrm{d}z \\ &\quad - \int_{z_0}^{z_N} (\lambda + 2\mu) \frac{\partial U_z}{\partial z}\frac{\partial \varphi}{\partial z}\mathrm{d}z \quad + [\tau_z \varphi]_{z_0}^{z_N} = 0, \end{aligned} \tag{A2}$$

where the boundary terms are:

$$\tau_r(z) = \mu \frac{\partial \bar{U}_r(z)}{\partial z} - k^2 \mu U_z(z), \tag{A3}$$

$$\tau_z(z) = (\lambda + 2\mu) \frac{\partial U_z(z)}{\partial z} + \lambda \bar{U}_r(z). \tag{A4}$$

Considering the following relation between wave potentials and displacements

$$U_r(z) = kp(z) + \frac{\partial q(z)}{\partial z}, \tag{A5}$$

$$U_z(z) = -\frac{\partial p(z)}{\partial z} - kq(z), \tag{A6}$$

then the boundary terms in eqs (A1) and (A2) can be expressed using wave potentials:

$$\tau_r(z) = 2k^2 \mu \frac{\partial p(z)}{\partial z} + \mu \frac{\partial^2 \bar{q}(z)}{\partial z^2} + k^2 \mu \bar{q}(z), \tag{A7}$$

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$$\tau_z(z) = k^2\lambda p(z) - (\lambda+2\mu)\frac{\partial^2 p(z)}{\partial z^2} - 2\mu\frac{\partial \bar{q}(z)}{\partial z}. \tag{A8}$$

At the deepest interface (z_N), the *P*- and *S*-wave potentials are determined using the Dirichlet boundary conditions:

$$p^{(N+1)}(z_N) = \frac{\alpha_{N+1}^2}{\omega^2}\left(\bar{U}_r^{(N+1)}(z_N) + \left.\frac{\partial U_z^{(N+1)}(z)}{\partial z}\right|_{z=z_N}\right), \tag{A9}$$

$$\bar{q}^{(N+1)}(z_N) = -\frac{\beta_{N+1}^2}{\omega^2}\left(k^2 U_z^{(N+1)}(z_N) + \left.\frac{\partial \bar{U}_r^{(N+1)}(z)}{\partial z}\right|_{z=z_N}\right). \tag{A10}$$

Based on the continuity of the displacements and the tractions at the interfaces, eqs (A9) and (A10) are modified as:

$$\rho_{N+1}\omega^2 p^{(N+1)}(z_N) - (\lambda_N+2\mu_N)\left.\frac{\partial U_z^{(N)}(z)}{\partial z}\right|_{z=z_N} - (\lambda_N+2\mu_{N+1})\bar{U}_r^{(N)}(z_N) = 0, \tag{A11}$$

$$-\rho_{N+1}\omega^2\bar{q}^{(N+1)}(z_N) = \mu_N\left.\frac{\partial \bar{U}_r^{(N)}(z)}{\partial z}\right|_{z=z_N} + (2\mu_{N+1}-\mu_N)k^2 U_z^{(N)}(z_N). \tag{A12}$$

Considering that no up-going waves exist in the bottom half-space, the *S*-wave potentials in the half-space layer can be expressed with exponential functions:

$$q^{(N+1)}(z) = q^{(N+1)}(z_N)\,e^{-i\gamma_{N+1}(z-z_N)} \qquad (z > z_N), \tag{A13}$$

where γ_{N+1} denotes the vertical wavenumber of *S* waves in the half-space. Then the derivatives of the *S*-wave potentials in eqs (A7) and (A8) can be written as

$$\frac{\partial \bar{q}^{(N+1)}(z)}{\partial z} = -i\gamma_{N+1}\bar{q}^{(N+1)}(z) \qquad (z > z_N). \tag{A14}$$

Since we only consider the surface and guided-*P* waves, the *P*-wave potentials in the half-space decay as the depth increases, which is hence able to be determined using the weak form

$$\int_{z_N}^{z_\infty}\rho\omega^2 p\varphi\mathrm{d}z - k^2\int_{z_N}^{z_\infty}(\lambda+2\mu)\,p\varphi\mathrm{d}z - \int_{z_N}^{z_\infty}(\lambda+2\mu)\frac{\partial p}{\partial z}\frac{\partial \varphi}{\partial z}\mathrm{d}z + \left[(\lambda+2\mu)\frac{\partial p}{\partial z}\varphi\right]_{z_N}^{z_\infty} = 0. \tag{A15}$$

The solution of eq. (A15) depends on the boundary conditions of *P*-wave potentials. The boundary conditions at z_N is determined using the Dirichlet boundary (A9) and the *P*-wave fields at z_∞ are simply set to be 0.

So far, all the governing equations and boundary conditions of the PSV wavefield have been expressed explicitly. By considering eqs (A1), (A2), (A7), (A8), (A11), (A12), (A15) and replacing k^2 with $\omega^2/\beta_{N+1}^2 - \gamma_{N+1}^2$, we have:

$$\int_{z_0}^{z_N}\left(\rho\omega^2 - \frac{\omega^2}{\beta_{N+1}^2}(\lambda+2\mu)\right)\bar{U}_r\varphi\mathrm{d}z - \int_{z_0}^{z_N}\mu\frac{\partial \bar{U}_r}{\partial z}\frac{\partial \varphi}{\partial z}\mathrm{d}z - \frac{\omega^2}{\beta_{N+1}^2}\int_{z_0}^{z_N}\left(\lambda\frac{\partial U_z}{\partial z}\varphi - \mu U_z\frac{\partial \varphi}{\partial z}\right)\mathrm{d}z + \gamma_{N+1}^2\left[\int_{z_0}^{z_N}(\lambda+2\mu)\,\bar{U}_r\varphi\mathrm{d}z + \int_{z_0}^{z_N}\left(\lambda\frac{\partial U_z}{\partial z}\varphi - \mu U_z\frac{\partial \varphi}{\partial z}\right)\mathrm{d}z\right] + \left[\begin{pmatrix}2\omega^2\rho_{N+1}\frac{\partial p^{(N+1)}(z)}{\partial z} + \omega^2\rho_{N+1}\bar{q}^{(N+1)}(z)\\ -2\gamma_{N+1}^2\mu_{N+1}\frac{\partial p^{(N+1)}(z)}{\partial z} - 2\gamma_{N+1}^2\mu_{N+1}\bar{q}^{(N+1)}(z)\end{pmatrix}\varphi\right]_{z_0}^{z_N} = 0, \tag{A16}$$

$$\int_{z_0}^{z_N}\left(\rho\omega^2 - \frac{\omega^2}{\beta_{N+1}^2}\mu\right)U_z\varphi\mathrm{d}z + \gamma_{N+1}^2\int_{z_0}^{z_N}\mu U_z\varphi\mathrm{d}z + \int_{z_0}^{z_N}\left(\mu\frac{\partial \bar{U}_r}{\partial z}\varphi - \lambda\bar{U}_r\frac{\partial \varphi}{\partial z}\right)\mathrm{d}z - \int_{z_0}^{z_N}(\lambda+2\mu)\frac{\partial U_z}{\partial z}\frac{\partial \varphi}{\partial z}\mathrm{d}z + \left[\begin{pmatrix}-\rho\omega^2 p^{(N+1)}(z) + 2\mu\gamma_{N+1}^2 p^{(N+1)}(z)\\ +2i\mu\gamma_{N+1}\bar{q}^{(N+1)}(z)\end{pmatrix}\varphi\right]_{z_0}^{z_N} = 0, \tag{A17}$$

$$\int_{z_N}^{z_\infty}\left(\rho\omega^2 - \frac{\omega^2(\lambda+2\mu)}{\beta_{N+1}^2}\right)p\varphi\mathrm{d}z + \gamma_{N+1}^2\int_{z_N}^{z_\infty}(\lambda+2\mu)\,p\varphi\mathrm{d}z - \int_{z_N}^{z_\infty}(\lambda+2\mu)\frac{\partial p}{\partial z}\frac{\partial \varphi}{\partial z}\mathrm{d}z = 0, \tag{A18}$$

$$\rho_{N+1}\omega^2 p^{(N+1)}(z_N) - (\lambda_N+2\mu_N)\left.\frac{\partial U_z^{(N)}(z)}{\partial z}\right|_{z=z_N} - (\lambda_N+2\mu_{N+1})\bar{U}_r^{(N)}(z_N) = 0, \tag{A19}$$

$$\rho_{N+1}\omega^2\bar{q}^{(N+1)}(z_N) + \frac{\omega^2}{\beta_{N+1}^2}(2\mu_{N+1}-\mu_N)U_z^{(N)}(z_N) + \mu_N\left.\frac{\partial \bar{U}_r^{(N)}(z)}{\partial z}\right|_{z=z_N} = \gamma_{N+1}^2(2\mu_{N+1}-\mu_N)U_z^{(N)}(z_N). \tag{A20}$$

The discretization of the upper governing equations is based on the spectral element method (SEM). Because the wavefield solutions of the above equations are horizontally analytical and vertically discrete, the method is called semi-analytical SEM (i.e. SASEM). Since SEM has been widely used in seismic problems, we do not devote space in this paper to introduce the basic theories and detailed implementation methods of SEM.

After discretization, eqs (A16) to (A20) results in a quadratic eigenvalue problem:

$$\omega^2\mathbf{M}\mathbf{U} - \mathbf{E}\mathbf{U} = \gamma_{N+1}^2\mathbf{K}_2\mathbf{U} + \gamma_{N+1}\mathbf{K}_1\mathbf{U}. \tag{A21}$$

To determine the eigenvalues and the right eigenvectors that satisfy eq. (A21), an order-reduction method is adopted. By in-

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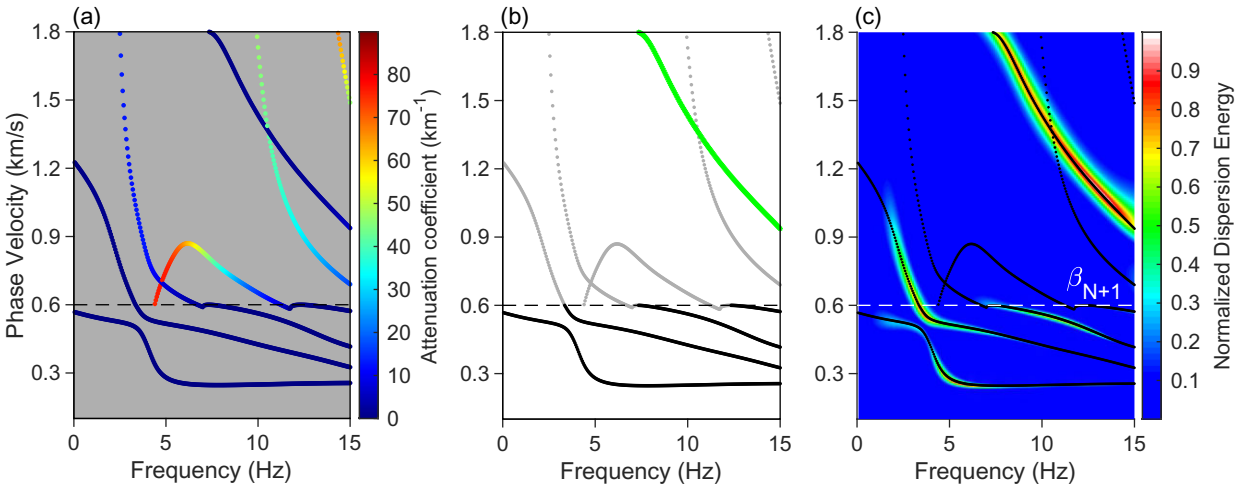


Figure C1. Dispersion for the LVL model at low frequencies. (a) Modal solutions from 0.05 to 15 Hz. (b) Modal classification results. Explanations of the markers and lines are identical to Fig. 4(c). (c) Theoretical dispersion spectrum for the vertical displacement.

troducing a new vector $\tilde{\mathbf{U}} = \gamma_{N+1}\mathbf{U}$, eq. (A21) can be rewritten as a generalized linear eigenvalue problem which can be easily solved:

$$\begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \omega^2\mathbf{M} - \mathrm{E} & -\mathbf{K}_1 \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \tilde{\mathbf{U}} \end{bmatrix} = \gamma_{N+1} \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_2 \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \tilde{\mathbf{U}} \end{bmatrix}. \tag{A22}$$

APPENDIX B CALCULATION OF THE LEFT EIGENVECTORS

It can be simply deduced that the two eigenvalue problems in eqs (7) and (11) have identical eigenvalues. To prove this, we first give a new equation by transposing all the terms in eq. (11):

$$\omega^2\mathbf{M}^T\mathbf{W}^T - \mathbf{E}^T\mathbf{W}^T = \gamma_{N+1}^2\mathbf{K}_2^T\mathbf{W}^T + \gamma_{N+1}\mathbf{K}_1^T\mathbf{W}^T. \tag{B1}$$

Each eigenvalue γ_{N+1} of eq. (7) must satisfy:

$$\left|\omega^2\mathbf{M} - \mathbf{E} - \gamma_{N+1}^2\mathbf{K}_2 - \gamma_{N+1}\mathbf{K}_1\right| = 0. \tag{B2}$$

Considering that the transposition of a matrix does not change the determinant, we have

$$\left|\omega^2\mathbf{M}^T - \mathbf{E}^T - \gamma_{N+1}^2\mathbf{K}_2^T - \gamma_{N+1}\mathbf{K}_1^T\right| = 0. \tag{B3}$$

Consequently, the eigenvalues of eq. (7) must satisfy eq. (11) as well. For any given eigenvalue γ_{N+1}, eq. (B1) becomes an underdetermined equation. Then $\mathbf{W}$ can be any one among the general solutions of eq. (B1). The general solutions of linear equations can be directly solved using mathematical software and libraries.

APPENDIX C DISPERSION ENERGY OF THE LVL MODEL AT LOW FREQUENCIES

To clearly show the low-frequency dispersion energy of the LVL model, the dispersion curves and TDS in the frequency range of 0.05 to 15 Hz are calculated (Fig. C1). It is noted that the fundamental and first normal modes exhibit proximity at frequencies around 3.5 Hz. Due to the remarkable dispersion energy of the first normal mode and its related apparent Σ leaky mode, we infer that mode kissing may occur in the low-resolution field data spectra for similar subsurface structures.

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